Getting Started in Research

EDP 612

Week 4

Plan for the Day

- Ethics Wrap Up
- Review remaining Ethics Slides
- Facebook Article Discussion
- Literature Matrix Group Discussion
- Getting Started in Research
- Preliminary Ideas Roundtable

Re: Ethics

Facebook Article

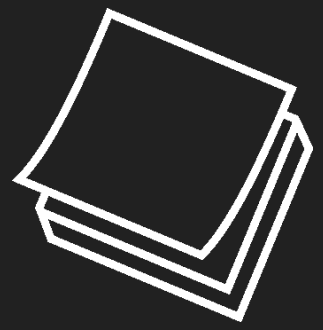
Literature Matrix Group Discussion

- In Groups, review your literature matrix.
- Compare/Contrast notes on articles
- Differences in organization?
- Use for your own Lit Review

Getting started designing a research project

Your own Ideas

- Take a few minutes and consider your own ideas for a research study as we go through the next set of slides:
  - Select a topic to study (maybe what you're going to do your lit review on)
  - Which broad approach do you plan to use (Qualitative, Quantitative)
  - Research Purpose Statement
  - 2-3 Research Questions or Hypotheses
  - Indented Measures
  - Sampling Method
- Prepare to share your initial ideas at the end of class!



Selecting a Research Topic

Where to get ideas?

- Literature
- Personal interests
- Observations

- Previous research experience

- Narrow your topic

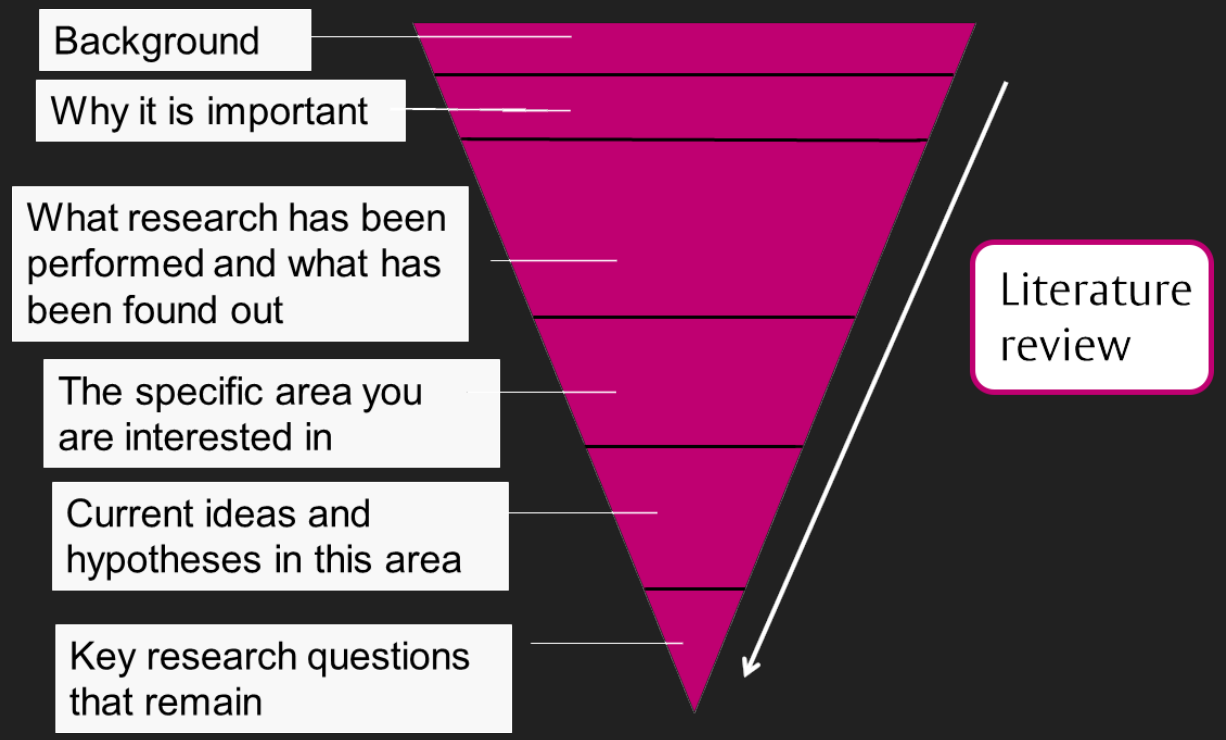
  - Undergraduate research
  - Mentor-Mentee Relationships in UR
  - Disciplinary knowledge transfer between mentors and mentees in UR

- Additional Considerations:

  - Researchable?
  - Significance, Value, Worth?
  - Has it been done before?

Lit Review

- Goal is to both review research in topic area AND to make it clear why your research questions can add to the field.
- Include seminal studies (landmark studies)
- Synthesize existing literature, do not summarize!



Research Purpose, Questions, and Hypotheses

Research Purpose Statement

- Purpose and objectives of the study.
    - May vary based on approach used (Qual verses Quant)

Hypotheses & Research Questions

- Hypothesis: predictive relationships between variables (quant)
    - Research Questions: guiding questions of the study
        - Quant research – can have both hypotheses and RQs but not always
        - Qual – Research questions may be more inductive or open ended.

Measurement & Variables

- Categorical Variables
  - Variables that are able to be sorted into meaningful groups
  - Student year in school, Race, Binaries (Yes, No; enrolled, not enrolled)
- Continuous Variables
  - Rely on progress and preserve meaningful distance between variables
  - Age, Income, time spent in condition
- Independent Variables
  - Influences or impacts other variables
- Dependent Variable
  - Affected or influenced by other variables
  - “depends” on other variables

Also...

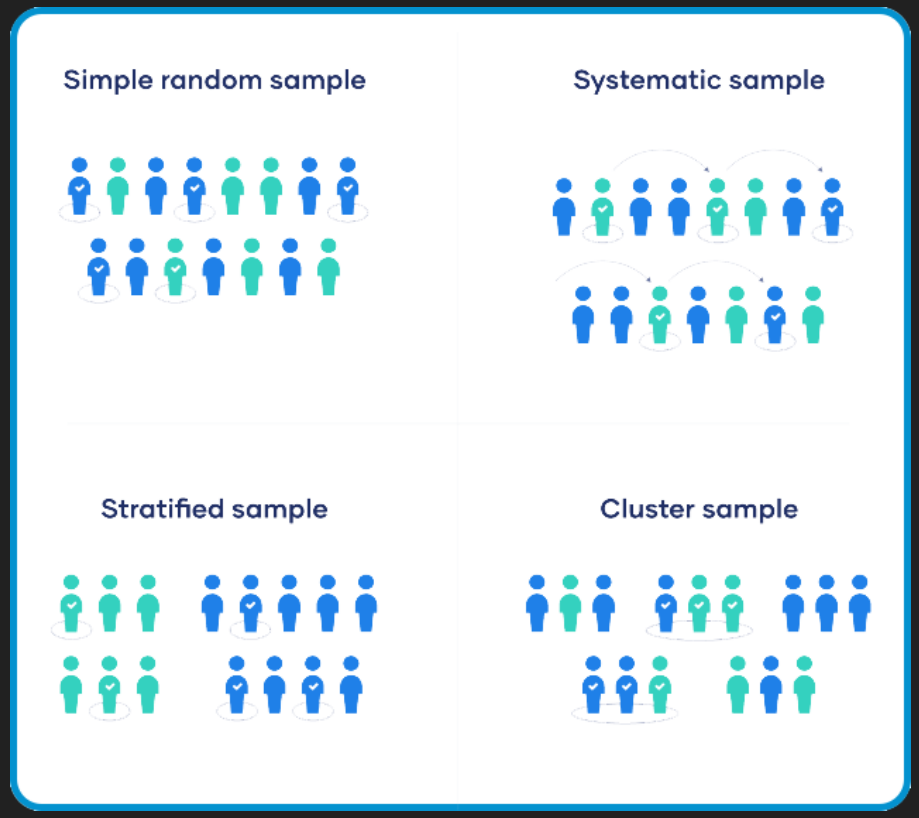
- Extraneous Variables
  - May impact data and aren’t controlled for in the study
  - Covariates
  - May impact the study, so you control for these in your study!

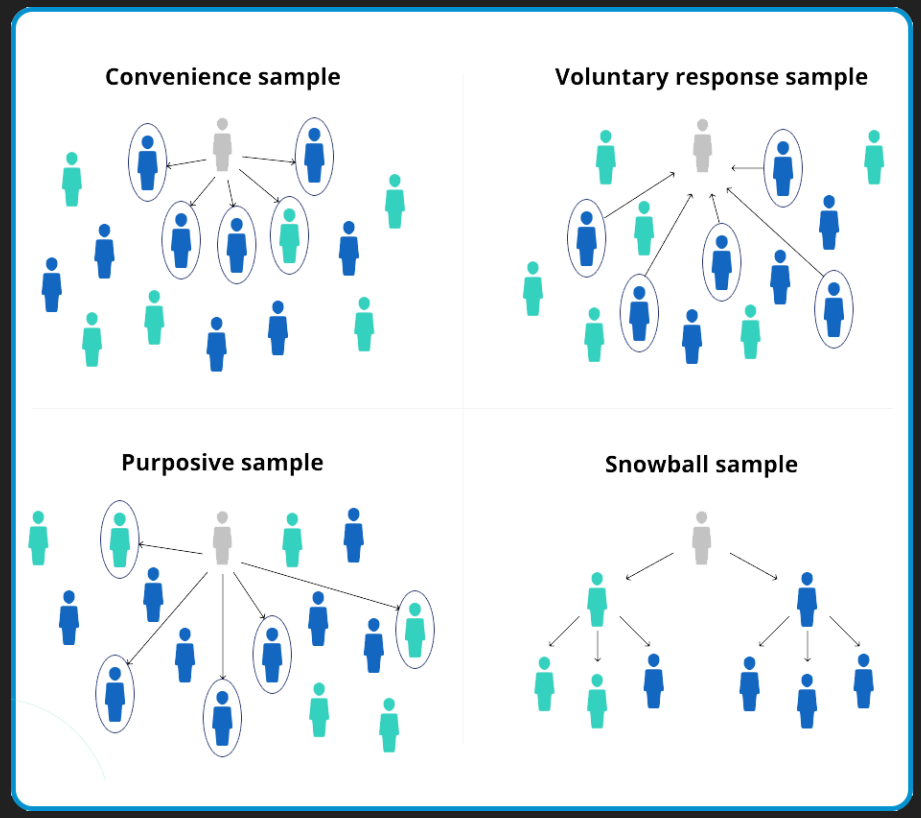
Sampling

- Things to consider:
  - Who is your population?
  - Who is the broad group you are interested in that you're pulling your sample from?
- Population: Undergraduate mentors and mentees at R1 institutions.
- Sample: 10 mentor-mentee pairs at R1 university on the mid-East coast.
  - Consider how many participants you might need:
  - Sample size calculators (quantitative research)
  - Reaching saturation (qualitative research) calculators

Sampling

- *Probability Sampling*: Participant selection based on strategy such that population has a known and nonzero chance of being selected. Typically used in quant research. Useful if you are hoping to generalize your findings.
- *Purposeful Sampling*: Selecting participants who are the best candidates for the study. Typically used in qualitative research!





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Roundtable

- Talk us through your ideas so far!
  - Study purpose, Hypotheses/Research Questions, Approach (qual or quant), Variables and measures of interest, Sampling
  - Ask each other (and us) questions!

For Next Week:

- Topic: Experimental and Quasi-Experimental Designs
  - Due: L1 - Selecting a Topic and Annotated Bib
    - (Similar to what we did in class today!)
    - Select a topic and describe it in 500 (+/- 50) words
    - Include an Appendix of the Annotated Bibliography
      - 5 articles in bibliography
    - Follow formatting guidelines (APA 7)
  - Read:
    - Leavy Chapter 4, p. 87-93.
    - Shadish , Cook, & Campbell, Chapter 1

Calendar of Tasks

L0	L0: Lit Review Process	Week 3
L1	L1: Topic Selection & Annotated Bib	Week 5
L2	L2: Annotated Bibliography	Week 7
L3	L3: Lit Review First Draft	Week 9
L4	L4: Lit Review Second Draft	Week 11
L5	L5: Lit Review Third Draft	Week 13
L6	L6: Lit Review Final Version!	Week 15 [1]

[1] This is the week of final exams.

That’s it!

Any questions?